

Tax Productivity and Development: A Quantitative Macroeconomic Analysis

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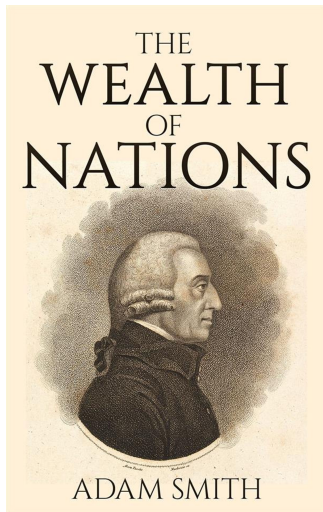
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The Importance of Tax Collection

- Core task of government; prerequisite for any public spending.
- Central to **'state capacity'** theory of development,

Besley and Persson (2011), Besley, Ilzetzki, and Persson (2013),
Mayshar, Moav, and Pascali (2022), Tilly (1990), others.

Of Taxes in *The Wealth of Nations*



*"Every tax ought to be so contrived as both to take out and to keep out of the pockets of the people as little as possible over and above what it brings into the public treasury of the state. . . **the cost of collecting it may be very considerable, and in the worst cases may amount to more than it brings in.**"*

Q: Why Was Egypt Great?



Egyptian Pyramids $\approx$ 3000 BCE



New Guinea sweet potatoes $\approx$ 5000 BCE

A: The Nilometer. Mayshar, Moav, Pascali (2021)'s Appropriability



Open Questions:

- Tax collection capacity is not easy to measure,
 - Typically proxied by Taxes/GDP , but quite indirect measure
 - Theories of state capacity $\Rightarrow$ tax capacity increases Taxes and GDP.
- Little consensus on *quantitative importance* of tax collection capacity,
 - Q: How much long-run GDP per capita increase if tax collection capacity in LICs raised?

Overview of This Paper

- **Data:** new data on tax collection across 100 countries
 - Tax collections (T) / collection eXpenditures (X)
 - Document robust & strong positive correlation with GDP per capita in:
 - (1) cross-section, (2) time series, and (3) across local states

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- **Toy model** of tax productivity
 - $\downarrow$ tax productivity $\Rightarrow$ $\downarrow$ public capital $\Rightarrow$ $\downarrow$ firm productivity $\Rightarrow$ $\downarrow$ GDP per capita
 - role of marginal and average productivity in tax collection
- **Quantitative analysis:** dynamic G.E. model to interpret data + simulate counterfactuals
 1. Tax administration in workhorse Ramsey model
 2. Public capital complimentary to private capital à la Barro (1990), baxter-King (1993)
 3. Reallocation to and from modern/traditional sector

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 - other macro aggregates

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 - new data tax collection data moments
 - other macro aggregates
- **A: +50% African LIC's tax productivity would boost LR GDP per capita by $\approx +4.7$ %**
 - **Order of magnitude larger** than simple “Hulten” logic, which suggests $+0.15$ %

Empirical Findings: Tax Collections / Expenditures

Measuring Tax Collections/ Expenditure

- Unit of Observation: tax administration

Measuring Tax Collection / Expenditure

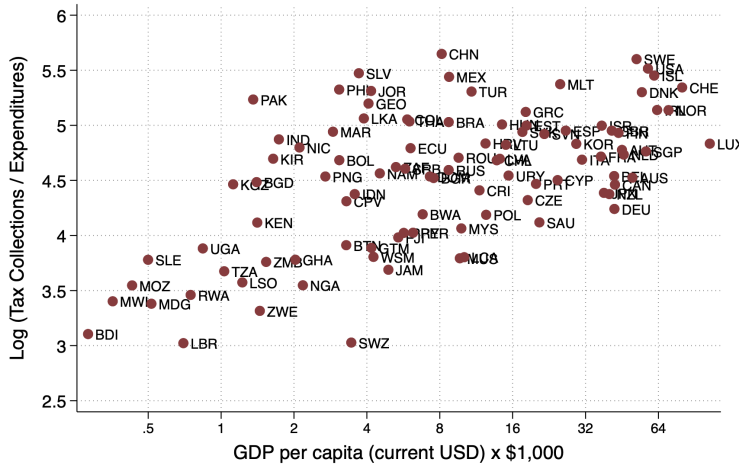
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Measuring Tax Collection / Expenditure

- Data Source
 - National tax authority documents
 - IMF Revenue Administration Fiscal Information Tool
- 100 countries, 625 country-year observations
- T = Tax Collections including PIT, CT, VAT, sales, and customs taxes
- X = Official tax collection authority budget
- Tax collection over collection expenditure, T/X
 1. Cross-country comparisons and regressions
 2. Historical Panel Regressions
 3. Sub-national states regressions

1. Tax Collection / Expenditure Increases with GDP per capita

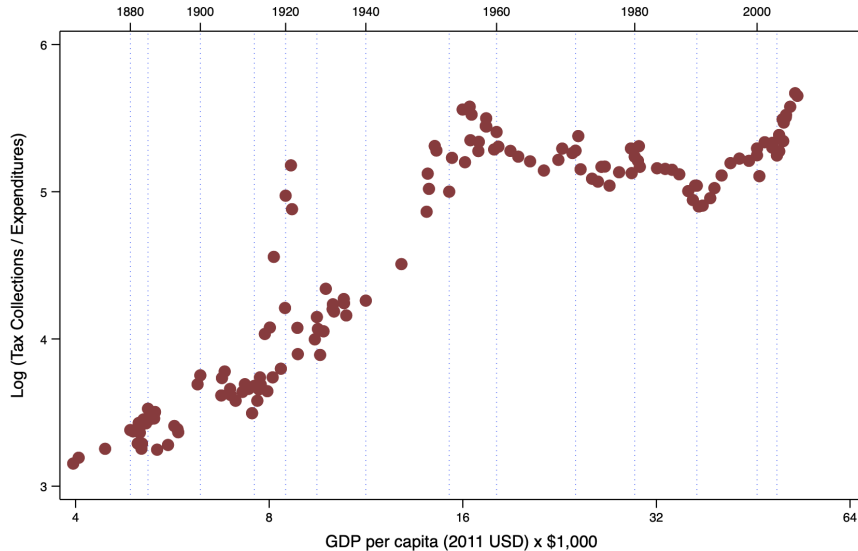


1. Descriptive Cross-Country Regressions: T/X and GDP Per Capita

Outcome	(1) Log T/X	(2) Log T/X	(3) Log T/X	(4) Log T/X	(5) Log T/X	(6) Log T/X	(7) Log T/X	(8) Log T/X
Log GDPpc	0.248*** (0.0389)	0.654*** (0.142)	0.225*** (0.061)	0.221*** (0.0586)	0.249*** (0.0395)	0.260*** (0.0398)	0.218*** (0.0484)	0.241*** (0.0399)
<i>N</i>	100	35	65	99	97	90	73	98
<i>R</i> ²	0.293	0.390	0.174	0.321	0.296	0.326	0.222	0.276
Modif. to column (1)		Only low, lower-middle inc. countries	Only upper- middle, high inc. countries	Cont. for PIT, VAT, CIT, tariff	Excl. if no VAT or sales tax	Excl. if no customs	Excl. if include soc. sec.	Excl. if resp for pop registry

Statutory Tax Rates

2. Tax Collection / Expenditure in the US: 1867 to 2017



2. Descriptive Panel Regressions over 16 Countries

	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for customs duties	(3) Log (T/X) Exclude if responsible for social security contributions	(4) Log (T/X) Exclude if changes in responsibilities
Log Real GDP pc	0.637*** (0.175)	0.636*** (0.197)	0.684*** (0.163)	0.644*** (0.193)
Observations	625	546	519	578
R-squared	0.563	0.541	0.574	0.541
Country FE	Yes	Yes	Yes	Yes

Australia (1972-2017); Bangladesh (1964-2015); Botswana (1971-2005); Canada (1924-2008); Denmark (1959-2006); India (1946-2016); Kenya (1962-1995); Malawi (1964-1996); Pakistan (1949-2005); South Korea (1987-2016); Sierra Leone (1964-1996); South Africa (1960-2007); Thailand (1959-2017); United Kingdom (1993-2017); United States (1877-2016); Uruguay (1980-2017).

3. Sub-National States Regressions

Outcome	(1) Log T/X	(2) Log T/X
Log GDP percap	0.567*** (0.143)	0.278*** (0.0606)
Sample	US states (1993-2017)	Indian states (1950-2017)
Number of states	50	28
State FE	Yes	Yes
N	850	964
R^2	0.719	0.777

(1) US states (1993-2016), (2) Indian states (1950-2017)

Takeaways

- **T/X positively correlated with GDP per capita**
 1. across countries, robust to tax admin purview & rates
 2. over time
 3. across subnational states

Takeaways

- **T/X positively correlated with GDP per capita**
 1. across countries, robust to tax admin purview & rates
 2. over time
 3. across subnational states
- Suggestive evidence, but need **structural model** to:
 - Formalize notion of **tax productivity**
 - Simulate effects of counterfactually raising tax productivity

A Toy Model of Tax Productivity

Environment

- Production linear in labor, normalized to 1, $Y = L = 1$.
- Tax rate $\tau \in [0, 1]$ finances a public good G .
- *Average* cost of raising one dollar of revenue $\mu(\tau)$.
- *Marginal cost of public funds (MCPF)* $\mu'(\tau)$.
- Consumption C , public goods G , and preferences U :

$$C = (1 - \mu(\tau)\tau)Y, \quad G = \tau Y \quad U(C, G) = C^\beta G^\alpha, \quad \beta, \alpha > 0.$$

- The planner maximizes

$$\max_{\tau \in [0, 1]} (1 - \mu(\tau)\tau)^\beta (\tau)^\alpha. \quad (1)$$

- Micro-found link between the tax collection technology and the cost of public funds

Constant returns in tax collection

- Tax administrators collect $AL_C = \tau L_Y$, $A > 0$ is tax productivity, $L_Y + L_C = L = 1$.
- Consumption public goods are $C = (1 - \tau)L_Y$, $G = \tau L_Y$.
- Solving for L_Y , planner's problem equiv to (1) with constant aver. cost of public funds μ :

$$\mu \approx 1 + \frac{\alpha + \beta}{\beta} \frac{1}{A}. \quad (2)$$

- optimal amount of taxed collected in (1) is $\tau = \frac{\alpha}{\mu(\alpha + \beta)}$.
- If μ is close to 1, taxes collected close frictionless tax collection, $\mu = 1$ and $\tau^* = \frac{\alpha}{(\alpha + \beta)}$
- With $A = 30$ (average T/X_{LIC}), $\beta = 2/3$, $\alpha = 1 - \beta$, $\mu = 1.05$, and $G = 0.31$.
- For high-income countries, $A \approx 200$, $\mu = 1.015$, i.e. $G = 0.33$.
- CRS tax collection can't generate meaningful differences in development across countries.

Decreasing returns in tax collection

- Suppose now $AL_C^\eta = \tau L_Y$, where $0 < \eta < 1$.
- Labor market clearing $L_Y + (\frac{\tau L_Y}{A})^{1/\eta} = 1 \implies$ variable cost of funds $\mu(\tau)$ with $\mu'(\tau) > 0$.
- FOC of planner choice of τ in (1)

$$\tau^2 = \frac{\alpha - \tau\mu(\tau)(\alpha + \beta)}{\beta\mu'(\tau)}. \quad (3)$$

- numerator is of the same order of magnitude across countries
- But, small differences in the MCPFs can lead to large differences in taxes collected.
- Credible quantification must model a DRS tax collection sector.

Quantitative Model of Tax Productivity and Development

Overview of Model Environment

- Three novel model features to quantify workhorse Ramsey model

Judd (1985), Chamley (1986), Straub and Werning (2020), Auclert, Cai, Rognlie, and Straub, (2024),

1. Tax administration in workhorse Ramsey model

- One input: tax collectors L_C
- Output: tax rate on modern sector wages, $T_t = G(A^C, L_{C,t})w_{M,t}L_{M,t}$
- Tax productivity A^C is exogenous, captures “tax collection capacity”

2. Public capital complimentary to private capital: $Y_M = F(A^M, k_G, k, L_M)$,

Musgrave (1959), Barro (1990), Baxter-King (1993), Weinzeirl (2018)
World Bank Long-Run Growth Model accounting for corruption

3. Reallocation to and from modern/traditional sector

Technology: Modern and Traditional Sectors

- **Modern Sector:** production function with public capital k_G , **taxed wages:**

$$Y_M = F(A^M, k_G, k, L_M),$$

hire private (wage $w_{M,t}$) and rent price capital (rate r_t)

$$\max_{k_t, L_{M,t}} F(A^M, k_{G,t}, k_t, L_{M,t}) - w_{M,t} L_{M,t} - r_t k_t.$$

- **Traditional Sector:** linear production function, **untaxed:**

$$Y_T = A^T L_T,$$

wage $w_T = A^T$.

Public Sector and Government

- **Tax Collection:** Taxes T_t are collected on modern sectors wages at rate $G(A^C, L_{C,t}) \leq 1$,

$$T_t = G(A^C, L_{C,t})w_{M,t}L_{M,t}.$$

- **Government budget:** can borrow to finance budget.

$$w_{M,t}L_{C,t} + R_t b_t + k_{t+1}^G \leq T_t + b_{t+1} + (1 - \delta^G)k_t^G + \omega_t(1 + r_t - \delta)k_t.$$

Public capital evolves as:

$$k_{G,t+1} = (1 - \delta^G) k_{G,t} + T_t.$$

Government cannot borrow to finance public capital.

Households

- Modern and traditional sector workers form a **representative household**

$$\max_{\{c_t, k_{t+1}, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t, L_{M,t}, L_{T,t})$$

- **Budget constraint**

$$c_t + k_{t+1} + b_{t+1} \leq (1 - \omega_t)(1 + r_t - \delta)k_t + (1 - \underbrace{G(A^C, L_{C,t})}_{\text{tax rate}})w_{M,t}L_{M,t} + w_T L_{T,t} + R_t b_t.$$

Initial private capital $k_0 = \bar{k}_0$.

Equilibrium

A competitive equilibrium is a set of allocations

$$\{c_t, L_{T,t}, L_{M,t}, L_{C,t}, k_t, k_{G,t}, b_t\}$$

and prices

$$\{w_{T,t}, w_{M,t}, r_t, R_t\},$$

such that:

1. Firm Optimality:

- Modern firms take prices and $k_{G,t}$ as given, and maximize profit.
- Traditional firms hire labor at wage $w_T = A^T$.

2. Household Optimality: The household maximizes utility for given prices and L_C .

3. Budget constraints: Government BC and public capital accumulation equations hold.

4. Market Clearing: The goods market clears:

$$c_t + k_{t+1} + \underbrace{w_{M,t} L_{C,t}}_{\text{collection personnel wages}} + k_{G,t+1} \leq (1 + r_t - \delta) k_t + w_{M,t} L_{M,t} + A^T L_{T,t} + (1 - \delta^G) k_{G,t}.$$

Planner's Problem

- Government chooses the sequence of personnel $\pi = \{L_{C,t}\}_{t=0,\dots,\infty}$ to maximize welfare:

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t u(c_t(\pi), L_{M,t}(\pi), L_{T,t}(\pi)),$$

subject to:

- the aggregate resource constraint,
- an **implementability condition** for consistency with household optimization,
- the government's budget constraint.

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Lemma

The implementability condition on the planner's allocation choice is:

$$\sum_{t=0}^{\infty} \beta^t \left[u_c(t) c_t - u_{L_M}(t) L_{M,t} - u_{L_T}(t) L_{T,t} \right] = u_c(0) R_0 k_0.$$

Optimal Tax Collection

At an interior steady state, the optimal tax collection choice satisfies:

$$\frac{1 - \beta(1 - \delta_G)}{G_{L_C}(A^C, L_C) L_M} = \underbrace{\frac{F_G + 1 - \delta_G}{(F_k + 1 + -\delta)(1 - \omega)} - 1}_{\text{net return of public capital relative to private capital}} + \underbrace{\beta \left(\frac{1 - \eta}{\eta} \right) F_{L_M} G L_C}_{\text{net benefit from wage increases - personnel pay}} \quad (4)$$

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Economic Interpretation:

- A high F_k (marginal return on private capital) favors private investment over public hiring.
- A high F_G (marginal return on public capital) enhances the attractiveness of public sector spending.
- A higher tax collection productivity A^C improves the effectiveness of tax collectors.
- A larger modern sector (L_M) expands the taxable base.
- A larger net marginal from wage increases, net of tax, personnel pay increases the government's incentive to hire tax collectors

All endogenous objects.

Quantitative Analysis

Quantitative Analysis

- Goal:
 - Calibration to a country like Ghana
 - Long-term effects of raising tax productivity in a country like Ghana

Functional Forms

- We set the following functional forms:

[diminishing marginal products] $G(A^C, L_C) = A^C L_C^\eta,$

[Barro-Baxter-King complementary] $F_M(k^G, k, L_M) = A_M(\theta k^G)^\zeta k^\mu L_M^{1-\mu},$

[linear traditional sector] $F_T(L_T) = A_T L_T,$

[CRRA] $u(c) = \frac{c^{1-\gamma}}{1-\gamma},$

[isoelastic disutility of labor] $u(L) = \frac{(L)^{1/\epsilon}}{1+1/\epsilon}.$

Experiment: Raise Productivity (A^C) of Ghana



See also Jawara, Knebelmann, Levine, Ndiaye, and Pouliquen (2025) [In the field. The Gambia]

Impact of Increase in Tax Productivity

ΔA_C	ΔGDP	ΔHulten	Δk	Δk_G	ΔTax	L_M/L	L_T/L	L_C/L	T/X
10	1.1	0.01	2.4	12.3	12.3	0.45	0.55	0.0012	41
20	2.1	0.02	4.3	24.4	24.4	0.45	0.54	0.0009	54
50	4.4	0.05	8.6	57.2	57.2	0.46	0.54	0.0004	155

Notes: Columns with Δ indicate percentage changes in A_C , GDP, Hulten predictor, capital (k), public capital (k_G), and tax revenue. Variables L_M/L , L_T/L , L_C/L and T/X show absolute levels. Baseline parameters are $\eta = 0.05$, $\zeta = 0.17$, $\theta = 0.58$, $A_M = 0.83$, $A_C = 0.13$, with a Domar weight of 0.00092637.

Conclusion and Future Work

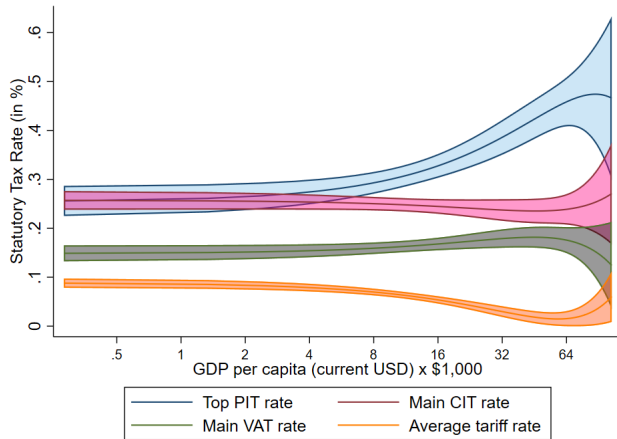
- New data and measures of tax collection capacity across countries
- Richest countries 30 more collection capacity than poorest
- Quantitative Ramsey model to study tax productivity and development
- **Long-run effects of tax productivity could be substantial**
 - much larger than those suggested by simple Hulten calculation
- Future work: fine-tuned calibrations to several countries, inference of tax productivity differences across countries, growth accounting, transition dynamics, alternative elasticities of substitution between public and private capital, etc.

Thanks!



Comments, criticism, suggestions: abdou.ndiaye@nyu.edu

Statutory Tax Rates in the Cross-Section of Countries



Notes: This figure plots the statutory tax rates against GDP per capita in the sample of 100 countries. The local polynomial for each type of tax rate is plotted, together with the 95 percent confidence interval.

Steady State

A steady state is determined by a system of 9 equations in 9 unknowns

$(w_M, w_T, r, L_M, L_T, L_C, k, k^G, c)$:

(i) Optimal collection:
$$\frac{1 - \beta(1 - \delta_G)}{G_{L_C}(A^C, L_C) L_M} = \beta \left[F_G + (1 - \delta_G) \right] - 1 + \beta \left(\frac{1 - \eta}{\eta} \right) F_{L_M C}$$

(ii) Private capital:
$$1 = \beta \left(F_k + 1 - \delta \right) + \beta \frac{1 - \eta}{\eta} F_{L_M, k} L_C,$$

(iii) Public capital:
$$k_G = \frac{1}{\delta_G} G(A^C, L_C) w_M L_M,$$

(iv) Return on private capital:
$$r = F_{M, k},$$

(v) Modern sector wage:
$$w_M = F_{M, L},$$

(vi) Traditional sector wage:
$$w_T = F_{T, L},$$

(vii) Modern labor supply:
$$u_{L_M} = u_c w_M (1 - G(A^C, L_C)),$$

(viii) Traditional labor supply:
$$u_{L_T} = u_c w_T,$$

(ix) Aggregate resource constraint:
$$c + k + w_M L_C + k_G = (1 + r - \delta)k + w_M L_M + A^T L_T.$$

Baseline Steady State

Variable	Parameter	Value
Modern labour	L_M	0.636
Traditional labour	L_T	0.773
Tax collectors	L_C	0.00192
Private capital	k	2.396
Public capital	k_G	0.755
GDP (output)	Y	1.484
Tax revenue	T	0.045
Average tax rate (%)	$\bar{\tau}$	9.51
Tax/GDP (%)	T/Y	3.05
Modern-sector share (%)	Y_M/Y	47.90

[Back to Main](#)

Transition Dynamics

Lemma

Assume that household preferences are separable in consumption and labor, with CRRA utility in consumption, and that the tax production function is linear in personnel

$G(A^C, L_C) = A^C L_C$. Then an interior steady state exists and the optimal personnel choice over the transition (for $t > 0$) satisfies:

$$\frac{(F_G(t+1) - \delta_G) - (F_k(t+1) - \delta)}{F_k(t+1) + 1 - \delta} = \frac{1}{A^C} \left(\frac{1}{L_{M,t}} - \frac{\beta(1 - \delta_G)}{L_{M,t+1}} \right).$$

Higher tax productivity A^C leads to higher levels of public capital (lower F_G). [Back to Main](#)